

**UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION**

Interconnection of Large Loads
to the Interstate Transmission System

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Docket No. RM26-4-000

**SUPPLEMENTAL COMMENTS OF ANTORA ENERGY
ON THE ADVANCE NOTICE OF PROPOSED RULEMAKING**

Antora Energy, Inc. (“Antora”) respectfully submits these supplemental comments in response to the Federal Energy Regulatory Commission’s (“FERC” or the “Commission”) Advance Notice of Proposed Rulemaking (“ANOPR”) on Large Load Interconnection.¹ Antora is a leading developer and manufacturer of thermal energy storage (“TES”) systems—highly flexible, low load-factor loads that charge with electricity approximately one-third of the time, using power only when generation exceeds demand, to deliver dispatchable, 24/7 industrial heat and power. As the Commission has recognized,² load flexibility can critically help to address the challenge of large load growth by increasing transmission and generation capacity utilization, putting downward pressure on rates, and by allowing the grid to accommodate new loads while infrastructure to serve them is being built.

As the Thermal Battery Alliance, of which Antora is a member, explained in its initial and reply comments in this docket,³ existing service models sharply constrain the commercial viability of flexible load development, and in particular development of highly flexible loads like TES, by treating them the same as inflexible loads. Since the end of the reply comment period,

¹ *Ensuring the Timely and Orderly Interconnection of Large Loads*, Advanced Notice of Proposed Rulemaking, Docket No. RM26-4-000 (Oct. 23, 2025) (ANOPR).

² *PJM Interconnection, LLC*, 193 FERC ¶ 61,217 (2025).; *see also* ANOPR at PP 23-24 (discussing principles for the Commission’s large load proceeding related to unauthorized injections or withdrawals and curtailment).

³ *See* Thermal Battery Alliance, Comments, Docket No. RM26-4-000 (filed Nov. 21, 2025); Thermal Battery Alliance, Reply Comments, Docket No. RM26-4-000 (filed Dec. 10, 2025).

certain RTOs/ISOs have proposed reforms to enable load flexibility. Antora applauds and supports these efforts. However, given that FERC is considering national solutions to large load interconnection,⁴ Antora submits these supplemental comments to provide Antora’s commercial experience with these proposals, including the particular solutions that hold the most promise and the ways they fall short of complete solutions to enable and unleash the benefits of load flexibility. Accordingly, Antora urges the Commission, in crafting its remedy in this proceeding, to go beyond these recent reforms to require all transmission providers to: (1) offer permanent non-firm transmission service; (2) at rates reflecting actual use that is consistent with cost causation principles; (3) with curtailment requirements for non-firm load integrated into the interconnection study process and resource adequacy requirements. These features are necessary for service frameworks to be workable for flexible loads, and therefore to ensure just and reasonable rates in the face of large load growth.

I. SPP’S EXPERIENCE DEMONSTRATES THE OPPORTUNITY FOR AND BARRIERS TO LOAD FLEXIBILITY

A. The Flexible Large Load Opportunity

Significant opportunities for flexibility exist that developers would take advantage of, were rules in place to appropriately reflect such flexibility in the interconnection process and transmission rates and services framework. As relates to Antora alone, Antora has identified a **6+ GW opportunity in SPP and 10+ GW nationally of immediately cost effective, viable, and highly-flexible thermal load opportunities based on current wholesale energy prices**, which can be sited at existing facilities in biofuels, chemicals, food processing, and agriculture.

⁴ See Georgina Mackey, *FERC Targets Data Center Interconnection Delays*, Broadband Breakfast (May 12, 2026), <https://broadbandbreakfast.com/ferc-targets-data-center-interconnection-delays/> (reporting that Chair Swett recently stated in remarks at the U.S. Energy Association: “We will be taking widespread action across the country in June” on large load interconnection.).

The system-wide benefits of enabling flexibility by Antora and other market participants would be substantial. Antora estimates 1 GW of flexible load projects could generate \$65.4 million in additional transmission revenues, reducing existing customer NITS fees by 1–12% depending on the zone in SPP. These loads increase utilization of existing generation and support price normalization during low-demand, high-generation periods when zero-marginal-cost generators have suppressed LMPs. Antora’s modeling of large-scale TES deployment in SPP shows system cost reductions of up to 3% in the mid case gas-price scenario and up to 8% in the high gas scenario, driven by lower wholesale costs, reduced capacity costs, and more efficient transmission utilization. These price mitigation effects present a critical tool to the Commission as it works to ensure the orderly integration of large loads onto the grid. But absent a framework that allows flexibility to be appropriately accounted for in interconnection and rates, customers like Antora will not pursue projects and these system cost reductions will not occur.

Highly flexible large loads face substantial interconnection barriers where, as in SPP, the RTO requires a one-to-one matching of non-coincident peak demand with new Designated Network Resources (DNRs). Highly flexible, curtailable large loads can avoid peaks and are inherently price sensitive, meaning they only operate when there is abundant low-cost energy generation available. An artificial DNR requirement is prohibitively expensive, unnecessary given the real-world operation of curtailable large loads like TES, and acts as a roadblock to new beneficial flexible loads. Without a realistic pathway to interconnection of flexible large loads, surplus energy that could be used by those loads goes unabsorbed—driving up curtailment and generator ramping, constraining grid utilization (thus increasing costs to existing customers of maintaining the grid), and eroding the viability of existing generation. SPP’s own data shows

January 2026 had a 4.6% increase in curtailed MWs and a 17% rise in constraint congestion month-over-month.

Antora's TES systems are designed to solve for this exact problem because of their unique model to connect and serve as large highly flexible and curtailable industrial loads. Other regions have regulatory models that better allow for connecting and serving large highly flexible, curtailable, industrial loads like Antora's TES systems. MISO allows large loads that do not increase the coincident peak to interconnect without new capacity requirements. SPP is developing the Price Adaptive Load Service for large price sensitive, flexible and curtailable loads to interconnect without DNRs or other capacity requirements. FERC should require all jurisdictional RTOs and transmission providers to move quickly to develop and implement similar viable interconnection pathways and affordable (based on cost causation) service for large, flexible and curtailable loads.

B. SPP's DNR Requirement Creates an Insurmountable Barrier

SPP requires all new loads to procure firm transmission and generation capacity sufficient to meet non-coincident peak demand—regardless of flexibility.⁵ By effectively assuming all loads operate continuously at full capacity, SPP's study process dramatically overstates what flexible loads actually require. As Antora has stated before the SPP Board, “low load-factor, price-sensitive, highly flexible loads have no current mechanism for gaining transmission access without Designated Network Resources that will not be used or needed.”

⁵ Southwest Power Pool, Inc./Open Access Transmission Tariff, Sixth Revised Volume No. 1, Definitions D, 1 Definitions D (8.0.10) (defining “Designated Resource” to require availability to “be called upon to meet the Transmission Customer’s load on a non-interruptible basis”); *see also id.* Attachment AQ — Delivery Point Assessment Process; *id.* Part III, section 29, Initiating Service & section 30, Network Resources; *id.* Attachment AX, Provisional Load Process.

This is particularly incompatible with highly flexible loads, such as TES, which operates at ~30–40% load factors that delivers 24/7 industrial heat while consuming energy only when there is excess generation available at the node of interconnection. Requiring such loads to procure generation capacity as though they consume around the clock yields unjust and unreasonable costs, contradicts cost-causation principles, and unduly discriminates against such loads by treating differently-situated customers the same.⁶

Other RTOs have recognized this. MISO allows LSEs to serve new off-peak load without a 1:1 DNR match, and removes Load Modifying Resources from capacity calculations due to interruptibility. This more flexible approach is one reason Antora’s first large scale commercial project, which we announced is in commissioning this week, is in MISO, and not SPP.

II. SPP’S RECENT EFFORTS TO ENCOURAGE FLEXIBLE LARGE LOADS CAN INFORM COMMISSION ACTION IN THIS DOCKET

As the Commission considers national action, it can draw on Antora’s commercial experience responding to SPP’s recent efforts to develop programs and service offerings intended to accommodate, recognize, or encourage flexible large loads. In particular, SPP has developed a Conditional High Impact Large Load Service (“CHILLS”),⁷ and is the process of developing a Price Adaptive Load Service (“PALS”). These products—and the significant commercial differences between them—should inform the Commission’s approach.

At a high level, Antora believes that SPP’s High Impact Large Load efforts should be recognized and celebrated. More than many other regions, SPP is taking action with the objective of facilitating large load interconnection in a timely way while lowering costs and maintaining

⁶ See, e.g., *Calpine Corp. v. PJM Interconnection, L.L.C.*, 163 FERC ¶ 61,236 at n.112 (2018) (“Typically, undue discrimination cases involve a seller charging a different rate to similarly situated customers; but undue discrimination can also occur when a seller charges the same rate to differently-situated customers.”).

⁷ See SPP, Filing, Docket No. ER26-1323-000 (filed Feb. 10, 2026).

reliability. However, Antora’s recent experiences with CHILLS demonstrates that it will not be sufficient to maximize the opportunity and benefits of large load flexibility. On the other hand, SPP’s development of PALS holds significantly more promise for customers like Antora. That promise has not yet been realized because of delays in moving this proposal forward, but the concepts being developed by SPP can nevertheless serve to inform the establishment of new non-firm service requirements, whether the Commission acts nationally or on a region-by-region basis.

A. SPP CHILLS Is Not a Sufficient Solution

Antora has engaged with SPP on the CHILLS since its inception. While we support the innovation to allow temporary non-firm transmission service, CHILLS remains fundamentally inadequate as a pathway for interconnection or service for highly flexible industrial loads like TES.

1. Temporary and Uncertain.

CHILLS is conditional and temporary, only available for a seven-year term, after which the customer must transition to firm service.⁸ During the CHILLS term, the customer must submit a transition plan to firm service, including identification of the specific generation asset that will be brought on-line to “serve” the new load. For flexible loads, the addition of this new generation resource to serve load is unnecessary and costly. CHILLS’ temporary structure mirrors the Interim NITS product that the Commission required PJM to establish in its recent Co-Location Order, but does not fill the additional need the Commission identified in that proceeding, for the transmission provider to offer a *permanent* non-firm transmission service option.⁹

⁸ *Id.* at 14.

⁹ See *PJM Interconnection, LLC*, 193 FERC ¶ 61,217 (2025).

2. Annual Reservations Inflate Costs to Prohibitive Levels.

In contrast to non-firm point-to-point transmission service, CHILLS requires annual reservations.¹⁰ This means that an Antora thermal storage site would have to pay for transmission at the peak and off peak non-firm point to point rates (which can be very expensive) despite the fact that it would only use power $\sim\frac{1}{3}$ of the time, and only during hours when LMPs are at their lowest, meaning the transmission system has excess generation at the point of interconnection, and the load should reduce transmission constraints, a net positive to the system. This approach is unjust and unreasonable, inconsistent with the Commission's cost causation principle, and unduly discriminatory for treating differently situated loads similarly—as well as prohibitively expensive for highly price sensitive industrial loads like thermal storage. Antora proposed an hourly utilization-based rate. This would preserve revenue from high load-factor customers while letting low load-factor loads pay only for the service they use, but this approach was not adopted for CHILLS.

B. SPP Price Adaptive Load Service (PALS) Provides the Right Framework

On the other hand, SPP is in the process of developing an approach that does hold the potential to enable flexible load customers like Antora: Price Adaptive Load Service (PALS). SPP staff are currently developing the PALS proposal and plan to bring it to the stakeholder process for approvals later this year. PALS would create a long-term, non-firm transmission service enabling flexible, price-sensitive loads to interconnect without matching capacity. It is a model that would deliver system-wide economic benefits and avoid costly overbuild of new generation and transmission. The framework rests on these principles:

¹⁰ SPP, Filing, Docket No. ER26-1323-000, at 15, 17 (filed Feb. 10, 2026).

- **Full Curtailability:** PALS loads are always dispatchable to zero. They are curtailed during reliability events, emergency conditions, and meaningful system peaks. SPP or the Transmission Owner can shut down PALS load immediately.
- **No Resource Adequacy Obligation:** Fully curtailable loads do not contribute to Planning Reserve Margin obligations and require no DNRs to interconnect.
- **Demand Bid Price Cap:** Locational Marginal Price caps in both the Day-Ahead and Real-Time markets prevent PALS loads from triggering higher-cost marginal units.
- **Non-Firm Transmission Charges:** Per-MWh billing at hourly non-firm rates ($\text{Zonal Annual Transmission Revenue Requirement} / \text{Transmission Zone Peak Load} / 8760$) reflect the true cost causation profile of curtailable loads; directly supporting NITS costs reduction for existing ratepayers. Foundationally, PALS customers are consuming power when the grid is in non-peak conditions, regardless of time of day, with the additional aspect of PALS being fully curtailable. Therefore, PALS should not be subject to arbitrarily defined peak rate periods based on times of the day.

III. FERC SHOULD ENSURE COMPARABLE PATHWAYS ACROSS ALL RTOS

The current interconnection paths in different ISOs/RTOs direct investment based on geography rather than grid need. To achieve the ANOPR's goals of ensuring timely and orderly interconnection of large loads, the Commission must reorient how grid operators plan for and serve large loads that can commit to flexibility. Accordingly, we recommend that the Commission in this proceeding direct each RTO/ISO to:

1. Offer a **permanent, long-term, non-firm transmission service** for fully-curtailable, price-responsive loads in addition to temporary or conditional service for loads wishing to move to firm. Recognizing that some customers such as Antora can limit withdrawals whether or not co-located with generation, this service should be available to any customer capable of meeting the necessary withdrawal limitations, whether or not co-located.
2. Set **transmission charges based on cost-causation principles** reflecting actual use that is consistent with how RTOs already bill other firm customers.
3. **Integrate** the non-firm, curtailable nature of flexible loads **into the interconnection study process** so assumptions reflect actual operating profiles.
4. **Eliminate DNR and resource adequacy requirements** that result in excessive costs when imposed on customers receiving fully curtailable service.
5. Study the **full range of benefits** that efficiently-sited highly flexible loads can provide to the system, including by avoiding the need for new transmission upgrades.

Together, these principles will facilitate achievement of the ANOPR's goals of enabling loads >20 MW to connect efficiently, fairly, and without discrimination.

IV. CONCLUSION

A massive wave of potentially flexible load is poised to interconnect, providing an opportunity to deliver billions in economic development, reduce curtailment, support existing generation, and lower transmission costs for ratepayers. This opportunity is being hamstrung by frameworks that assume all loads operate at full capacity around the clock.

In SPP, Antora has been working for more than two years to establish a different pathway that enable flexible industrial loads to benefit the grid. PALS offers a sound model—protecting reliability through full curtailability, limiting price impacts through a bid cap, avoiding resource adequacy complications by design, and charging based on actual use. It must be completed promptly and on reasonable terms. And the principles that make PALS workable should not be confined to SPP—every jurisdictional RTO and transmission owner should be required to offer comparable, permanent, cost-causation-based non-firm service for flexible loads.

However the Commission acts in this docket, Antora urges action that ensures flexible loads can connect on terms that reflect their actual use—promptly, permanently, and at costs grounded in cost causation. In so doing, the Commission can ensure that transmission providers are making the most of the transmission system in a manner that facilitates the timely, orderly, and affordable interconnection of large loads, as the ANOPR directed.

Respectfully submitted,

/s/ Noah Long
Noah Long
Director, State and Regulatory Affairs
Antora Energy, Inc.
200 West De Vargas St
Santa Fe, NM 87501
noah.long@antora.com